

PRODUCT · DC-AIR®

DC-Air®

True Wireless® Intraoral X-Ray Sensor

DC-Air® is Athlos's premium wireless intraoral X-ray sensor. A single-crystal silicon / CMOS direct-conversion detector with 26 µm pixels, 16-bit grayscale and 1.25 megapixel resolution, paired with BLE 5.0 True Wireless® operation and dynamic per-exposure self-calibration.

Also known as DCiS / Athlos-Air® / Athlos-1



1 Key highlights

DETECTOR Single-crystal silicon / CMOS	PIXEL PITCH 26 µm	ACTIVE AREA 35.1 × 24.7 mm
RESOLUTION 1,249,920 px · 16-bit	WIRELESS Bluetooth LE 5.0 · up to 3 m	BATTERY Li-ion · up to 10 h / 300 X-rays

2 Kit contents

SENSOR · WIOS-S2 DC-Air® Intraoral · universal size	CHARGING · WIOS-DS2 Docking station Magnetic · USB-C · charge & pair	POSITIONING Zero Profile® holders Anterior · BW · posterior	INTEGRATION TWAIN driver Single-layer · Athlos SDK (C++)
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3 Technical specifications

Detector & image3.1				Imaging performance & exposure3.2			
Detector	Single-crystal direct conversion	Si / CMOS	Silicon-based intraoral sensor	MTF @ 5 lp/mm	> 70	%	Image Accuracy (IA)
Pixel physical size	26	µm	Square pixel	MTF @ 10 lp/mm	> 40	%	High-frequency contrast retention
Active area	35.1 × 24.7	mm²	Universal size — anterior, BW, posterior	Exposure time	0.05 – 0.5	s	Compatible with standard intraoral X-ray sources
Number of pixels	1,249,920	px	≈ 1.25 megapixel	Tube voltage	60 – 70	kV	Recommended operating range
Grayscale depth	16	bit	65,536 gray levels	Tube current	2 – 8	mA	Recommended operating range
Sensor mode of operation	Global shutter	—	Whole frame captured simultaneously	Trigger to display	≈ 6	s	From X-ray trigger to image on PC
Triggering	Automatic on X-ray start	—	No external trigger wiring	Onboard image storage	1 image	—	Safety buffer until handoff
Self-calibration	Dynamic · per capture	—	Consistent quality with every exposure	Patient population	General	—	Adult & pediatric intraoral exams

4 Clinical applications

Intraoral radiography - Periapical imaging — anterior & posterior - Bitewing imaging — caries detection & interproximal - Full-mouth series (FMX) - Endodontic working-length & treatment verification	Clinical settings - General dental practice — replaces wired sensors & phosphor plates - Multi-operatory clinics — one sensor moves between rooms - Specialist practice — endodontics, prosthodontics, periodontics - Premium / boutique clinics prioritising patient experience
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Technical specifications (continued)

Wireless connectivity3.3				Battery & docking station3.4			
Radio	Bluetooth® Low Energy	BLE 5.0	True Wireless® connectivity	Sensor battery type	Lithium-Ion	–	Rechargeable, integrated
Operating range	up to 3	m	Sensor to docking station	Battery capacity	32	mAh	
BLE Rx sensitivity	–89	dBm		Recharge cycle	up to 10	h / 300 X-rays	Typical clinical day
Trigger to display time	≈ 6	s	From X-ray trigger to on-screen image	Battery gauge	Yes	–	Charge state visible in software
BLE location services	Yes	–	For device finding (while powered)	Docking	Magnetic	–	For robustness and ease of use
Onboard storage	1 image	–	Safety buffer	Dock input voltage	+5 ± 10%	V DC	USB-powered
				Dock input power (max)	2.5	W	

Environmental & enclosure3.5				Interfaces3.6			
Operating temperature	+10 to +40	°C	Ambient	Wireless interface	BLE 5.0	–	Sensor → docking station
Transportation temperature	–20 to +50	°C		Wired interface	USB 2.0 High Speed	–	Docking station → PC
Storage temperature (long term)	0 to +40	°C		Connector (docking station)	USB-C	–	
Humidity (non-condensing)	30 – 95	%RH		Cable (dock to PC)	USB-C → USB Type A	–	Standard supplied
IP protection level	Enhanced IP67	–	Dust-tight · water-resistant	Triggering	Automatic	–	On X-ray start — no external trigger wiring
Encapsulation	Ultrasonic bonded	–	Robust seal, no glue lines	Accelerometer	N/A	–	Not equipped
Encapsulation material	PPSU	–	Hydrolytic stability & toughness				
Biocompatibility	ISO 10993	–	Agency rating				

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Software & compatibility

End-user software4.1				OEM integration4.2			
Interface	Optimised single-layer TWAIN	–	Driver included	Athlos SDK	Included	–	Source-code examples for integration
Acquisition workflow	Direct to PMS	–	No intermediate viewer required	SDK language	C++	–	Calling-function definitions
Compatible imaging software: Dentrix Ascend · Eaglesoft · Open Dental · Dexis · TigerView · Romexis · Acteon · Apteryx · tab32 · MOGO · Dovetail · Curve Dental · XDR Radiology · TDO · VistaSoft · Theia · DentalEye · ArchiMED suite — plus most TWAIN-enabled software.				Image processing	Locally adaptive filters	–	Advanced, included
				Image output	Raw / processed	–	16-bit grayscale

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Ordering & configuration

Order codes5.1			
DC-Air® sensor	WIOS-S2	universal	Wireless Intraoral Sensor
Docking station	WIOS-DS2	–	Magnetic dock, USB-C
DC-Air® standard kit	WIOS-S2 + WIOS-DS2	kit	Sensor · dock · Zero Profile® holders · TWAIN driver

DOCUMENT	REVISION	STATUS	ORIGIN
DCAIR-DS-2026.1	A · May 2026	Production release	Athlos Oy · Finland

COMPANION

For the commercial narrative, clinical workflow imagery and the full DC-Air® award history (Dental Product Shopper, Clinicians Report, Dental Advisor 2024/2025), see the **DC-Air® Brochure**.

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Contact

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